

15. 设 \$(X, Y)\$ 服从区域 \$D = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq 1\}\$ 上的均匀分布, 求相关系数 \$\rho_{XY}\$.

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{DX} \sqrt{DY}} \quad \text{Cov}(X, Y) = E(XY) - EX \cdot EY \quad \rho_{XY} = \frac{1}{5} = 1$$

$$E(XY) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xy \cdot p(x, y) dx dy = \int_0^1 \int_0^1 xy dx dy$$

$$= \int_0^1 x dx \int_0^1 y dy = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

$$EX = \int_0^1 x \cdot \frac{1}{4} dx = \frac{1}{8}, \quad EY = \int_0^1 y \cdot \frac{1}{4} dy = \frac{1}{8}$$

$$DX = EX^2 - (EX)^2 = \int_0^1 x^2 \cdot \frac{1}{4} dx - \left(\frac{1}{8}\right)^2 = \frac{1}{12} - \frac{1}{64} = \frac{5}{192}$$

$$DY = \frac{5}{12}$$

$$\rho_{XY} = \frac{1/5}{\sqrt{5/192} \sqrt{5/12}} = 0$$

6. $\text{Cov}(X, Y) = EXY - EX \cdot EY$

$$= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xy \cdot p(x, y) dx dy = \frac{5}{144}$$

$$= \int_0^1 dx \int_0^1 xy (2-x-y) dy$$

$$= \int_0^1 dx \left[xy^2 - \frac{1}{2}xy^3 - \frac{1}{2}xy^3 \right]_0^1$$

$$= \int_0^1 x \cdot \frac{1}{3}x^2 - \frac{1}{6}x^3 dx = \int_0^1 \frac{2}{3}x^2 - \frac{1}{6}x^3 dx$$

$$= \left[\frac{2}{9}x^3 - \frac{1}{24}x^4 \right]_0^1 = \frac{2}{9} - \frac{1}{24} = \frac{5}{24}$$

$$EX^2 = \int_0^1 x^2 \cdot \left(\frac{3}{2}-x\right) dx = \left[\frac{1}{3}x^3 - \frac{1}{2}x^4 \right]_0^1 = \frac{1}{3} - \frac{1}{2} = \frac{1}{6}$$

$$EX = \int_0^1 x \cdot \left(\frac{3}{2}-x\right) dx = \left[\frac{3}{4}x^2 - \frac{1}{2}x^3 \right]_0^1 = \frac{3}{4} - \frac{1}{2} = \frac{1}{4}$$

$$DX = EX^2 - (EX)^2 = \frac{1}{6} - \left(\frac{1}{4}\right)^2 = \frac{5}{24}$$

$$EY^2 = \int_0^1 y^2 \cdot \left(\frac{3}{2}-y\right) dy = \frac{1}{6}, \quad EY = \frac{1}{4}, \quad DY = \frac{5}{24}$$

$$\rho_{XY} = \frac{\frac{5}{24}}{\sqrt{\frac{5}{24}} \sqrt{\frac{5}{24}}} = 1$$

$$\rho_{XY} = \frac{11}{144} = \frac{11}{144}$$

$$\text{Cov}(X, Y) = \frac{11}{144} \approx 0.0764$$

17. 已知随机变量 \$X, Y\$ 的相关系数

$$\rho = \frac{EXY - EX \cdot EY}{\sqrt{DX} \sqrt{DY}}$$

$$EXY = E[ax+by+cx+dy] = E[acx^2 + bc^2 + adx + bdy]$$

$$EX \cdot EY = (aEX+b)(cEX+d) = acEX^2 + bcEX + adEX + bd$$

$$= acEX^2 + bcEX + adEX + bd$$

$$EXY - EX \cdot EY = ac[EXY - EX^2]$$

$$DX = D(ax+b) = a^2DX, \quad DY = D(cy+d) = c^2DY$$

$$\rho' = \frac{EXY - EX \cdot EY}{\sqrt{DX} \sqrt{DY}} = \frac{ac[EXY - EX^2]}{a^2c \sqrt{DX} \sqrt{DY}} = \frac{EXY - EX^2}{a^2c \sqrt{DX} \sqrt{DY}} = \rho$$

18. 计算 \$X, Y\$ 的相关系数, 并判断是否独立

$$EXY = \sum_{i=1}^n x_i y_i p(x_i, y_i) = \frac{3}{8} \cdot \frac{1}{8} \cdot \frac{1}{8} = \frac{3}{512}$$

$$EX = \sum_{i=1}^n x_i p(x_i) = \frac{3}{8} \cdot \frac{1}{8} = \frac{3}{64}$$

$$EXY - EX \cdot EY = 0$$

$$\rho = 0$$

B组 Chapter 4 随机变量的数学期望

1. $P(X=k) = \frac{1}{2^k}, k=1, 2, \dots$, 求 EX

$$EX = \sum_{k=1}^{\infty} k P(X=k) = \sum_{k=1}^{\infty} k \cdot \frac{1}{2^k}$$

$$\frac{1}{2} EX = \sum_{k=1}^{\infty} \frac{k}{2^{k+1}} = \frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \dots + \frac{k-1}{2^k} + \frac{k}{2^{k+1}}$$

$$\frac{1}{2} EX = EX - \frac{1}{2} EX = \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots + \frac{1}{2^k} + \frac{1}{2^{k+1}}$$

$$EX = 2 + \frac{1}{2^{k-1}} = 2 + \frac{n-2}{2^n}, n \rightarrow \infty \text{ 时 } EX = 2$$

2. 从数 0, 1, ..., n 中任取两个不同的数 X, Y, 求这两个数差的绝对值的期望

$$X \text{ 或 } Y = 0, \frac{n+1}{C_{n+1}^2}$$

$$X \text{ 或 } Y = 1, \frac{n}{C_{n+1}^2}$$

$$X \text{ 或 } Y = 2, \frac{n-1}{C_{n+1}^2}$$

$$\dots, \frac{1}{C_{n+1}^2}$$

$$\sum_{k=0}^n k^2 \text{ 推导}$$

$$(k+1)^3 - k^3 = 3k^2 + 3k + 1$$

$$\sum_{k=0}^n [(k+1)^3 - k^3] = 3 \sum_{k=0}^n k^2 + 3 \sum_{k=0}^n k + (n+1)$$

$$1^3 - 0^3 + 2^3 - 1^3 + \dots + (n+1)^3 - n^3 = 3S_0 + 3 \frac{n(n+1)}{2} + n+1$$

$$S_0 = \frac{(n+1)^3 - \frac{3}{2}n^2 - \frac{3}{2}n - n - 1}{3} = \frac{n^3 + \frac{3}{2}n^2 + \frac{3}{2}n - 1}{3} = \frac{n(n+1)(n+2)}{6}$$

3. 对目标连续射击直到命中为止, 每次命中且击中目标的概率为 p, 试求子弹的平均消耗量

$$P(X=k) = C_{n-1}^{k-1} p^{k-1} (1-p)^{n-k} p = \frac{d}{dp} \left[\sum_{k=1}^n (1-p)^{k-1} p^k \right] / dp$$

$$EX = \sum_{k=1}^n k \cdot C_{n-1}^{k-1} p^{k-1} (1-p)^{n-k} p = -1 + (1-p)^{n+1} / (1-p)$$

$$EX = \sum_{k=0}^n \frac{k!}{(n-1)!(k-n)!} p^n (1-p)^{k-n} = \sum_{k=0}^n \frac{k! \cdot n}{n!(k-n)!} p^n (1-p)^{k-n}$$

$$= \sum_{k=0}^n n \binom{k}{n} p^n (1-p)^{k-n} = np^n \sum_{k=0}^n \binom{k}{n} (1-p)^{k-n}$$

$$= np^n \left[1 + \sum_{j=1}^n \binom{j}{n} (1-p)^j \right] = np^n \left[1 + \sum_{j=1}^n \frac{j!}{n!(j-n)!} (1-p)^j \right]$$

$$\text{广义的二项式展开定理 } (1-x)^{-r} = \sum_{j=0}^{\infty} \binom{r+j-1}{j} x^j$$

$$\text{利用递推公式 } \binom{n}{k} = \frac{n}{k} \binom{n-1}{k-1} \text{ 所以 } \binom{j+n}{n} = \binom{j+n-1}{n-1}$$

$$\text{和变为 } \sum_{j=0}^n \binom{j+n}{n} (1-p)^j = (1-p)^{-n-1} \cdot (1-p)^{n+1} = 1$$

$$\text{因此 } \sum_{j=0}^n \binom{j+n}{n} (1-p)^j = (1-p)^{-n-1} \cdot (1-p)^{n+1} = 1$$

$$\text{设 } n \text{ 次击中目标子弹的消耗量为 } X, \text{ 则 } X \text{ 等于每次击中目标子弹消耗之和}$$

$$X_i \text{ 表示第 } i \text{ 次击中目标子弹消耗的子数, 那么 } X = X_1 + X_2 + \dots + X_n$$

$$\text{而 } X_i, i=1, 2, \dots, n \text{ 服从几何分布, 其分布为 } P(X_i=k) = (1-p)^{k-1} p, k=1, 2, \dots$$

$$\text{其中 } p=1-p, \text{ 所以 } EX_i = \frac{1}{1-p}, \text{ 从而 } EX = EX_1 + EX_2 + \dots + EX_n = \frac{n}{1-p}$$

4. 设 $X \sim B(n, p)$, 求 EX, DX

$$P(X=k) = C_n^k p^k (1-p)^{n-k}$$

$$EX = \sum_{k=0}^n k C_n^k p^k (1-p)^{n-k}$$

$$= \sum_{k=1}^n k C_n^k p^k (1-p)^{n-k}$$

$$= \sum_{k=1}^n \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k} = \sum_{k=1}^n \frac{n!}{(k-1)!(n-k)!} p^k (1-p)^{n-k}$$

$$= n p \sum_{k=1}^n \frac{(n-1)!}{(k-1)!(n-k)!} p^{k-1} (1-p)^{(n-1)-(k-1)}$$

$$= n p (p + 1-p)^{n-1} = np$$

$$DX = EX^2 - (EX)^2$$

$$EX^2 = \sum_{k=0}^n k^2 C_n^k p^k (1-p)^{n-k}$$

$$= \sum_{k=0}^n (k(k-1) + k) C_n^k p^k (1-p)^{n-k}$$

$$= \sum_{k=2}^n \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k} + \sum_{k=1}^n k C_n^k p^k (1-p)^{n-k}$$

$$= \frac{n(n-1)}{2} p^2 (1-p)^{n-2} + np$$

$$= \frac{n(n-1)}{2} p^2 (1-p)^{n-2} + np$$

$$DX = EX^2 - (EX)^2 = \frac{n(n-1)}{2} p^2 (1-p)^{n-2} + np - n^2 p^2$$

$$= np - np^2 = np(1-p)$$

5. $\square D$

$$P(X=V_1) = P_1 P_2 \quad EX_1 = P(X=V_1)V_1 + P(X=V_2)V_2 = P_1 P_2 V_1 + P_1 P_2 V_2$$

$$P(X=V_2) = P_1 P_2 \quad EX_2 = P(X=V_2)V_2 + P(X=V_1)V_1 = P_1 P_2 V_2 + P_1 P_2 V_1$$

$$P(X=V_1+V_2) = P_1 P_2 \quad EX_3 = P(X=V_1+V_2)(V_1+V_2) = P_1 P_2 (V_1+V_2)$$

$$EX_1 - EX_2 = P_1 P_2 V_1 - P_1 P_2 V_2 = P_1 P_2 (V_1 - V_2)$$

$$= P_1 P_2 (V_1 - V_2) \quad \text{若 } P_1 P_2 V_1 > P_1 P_2 V_2 \quad \text{则 } P_1 P_2 > 0$$

$$= P_1 P_2 (V_1 - V_2) \quad \text{若 } P_1 P_2 V_1 < P_1 P_2 V_2 \quad \text{则 } P_1 P_2 < 0$$

$$P(X,Y) = P_1 P_2 \quad EX_1 = P_1 P_2 V_1 \quad EX_2 = P_1 P_2 V_2$$

6. 设 X 服从正态分布 $N(\mu, \sigma^2)$, 求 $E[X^k]$

$$E[X^k] = \int_{-\infty}^{+\infty} x^k \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

$$= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{+\infty} (y+\mu)^k e^{-\frac{y^2}{2\sigma^2}} dy$$

$$= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{+\infty} y^k e^{-\frac{y^2}{2\sigma^2}} dy + \mu \int_{-\infty}^{+\infty} y^{k-1} e^{-\frac{y^2}{2\sigma^2}} dy$$

$$I_k = \int_{-\infty}^{+\infty} y^k e^{-\frac{y^2}{2\sigma^2}} dy$$

$$I_k = -\int_{-\infty}^{+\infty} y^{k-1} d e^{-\frac{y^2}{2\sigma^2}}$$

$$= -e^{-\frac{y^2}{2\sigma^2}} y^{k-1} \Big|_{-\infty}^{+\infty} + \frac{k-1}{\sigma^2} \int_{-\infty}^{+\infty} y^{k-2} e^{-\frac{y^2}{2\sigma^2}} dy$$

$$I_k = \frac{k-1}{\sigma^2} I_{k-2}$$

$$I_0 = \int_{-\infty}^{+\infty} e^{-\frac{y^2}{2\sigma^2}} dy = \sqrt{2\pi}\sigma$$

$$I_2 = \frac{1}{\sigma^2} I_0 = \sqrt{2\pi}\sigma$$

$$I_4 = \frac{3}{\sigma^4} I_2 = \frac{3}{\sigma^4} \sqrt{2\pi}\sigma$$

$$I_6 = \frac{15}{\sigma^6} I_4 = \frac{15}{\sigma^6} \frac{3}{\sigma^4} \sqrt{2\pi}\sigma$$

$$I_8 = \frac{105}{\sigma^8} I_6 = \frac{105}{\sigma^8} \frac{15}{\sigma^6} \frac{3}{\sigma^4} \sqrt{2\pi}\sigma$$

$$I_{2k} = \frac{(2k-1)!!}{\sigma^{2k}} \sqrt{2\pi}\sigma$$

$$E[X^{2k}] = \frac{(2k-1)!!}{\sigma^{2k}} \sqrt{2\pi}\sigma$$

$$= \frac{(2k-1)!!}{\sigma^{2k-1}}$$

$$E[X^{2k+1}] = 0$$

$$E[X^{2k}] = \frac{(2k-1)!!}{\sigma^{2k-1}}$$

$$E[X^{2k+1}] = 0$$

$$E[X^{2k}] = \frac{(2k-1)!!}{\sigma^{2k-1}}$$

$$E[X^{2k+1}] = 0$$

$$E[X^{2k}] = \frac{(2k-1)!!}{\sigma^{2k-1}}$$

$$E[X^{2k+1}] = 0$$

$$E[X^{2k}] = \frac{(2k-1)!!}{\sigma^{2k-1}}$$

$$E[X^{2k+1}] = 0$$

$$E[X^{2k}] = \frac{(2k-1)!!}{\sigma^{2k-1}}$$

$$E[X^{2k+1}] = 0$$

$$E[X^{2k}] = \frac{(2k-1)!!}{\sigma^{2k-1}}$$

$$E[X^{2k+1}] = 0$$

$$E[X^{2k}] = \frac{(2k-1)!!}{\sigma^{2k-1}}$$

$$E[X^{2k+1}] = 0$$

$$E[X^{2k}] = \frac{(2k-1)!!}{\sigma^{2k-1}}$$

$$E[X^{2k+1}] = 0$$

$$E[X^{2k}] = \frac{(2k-1)!!}{\sigma^{2k-1}}$$

$$E[X^{2k+1}] = 0$$

$$E[X^{2k}] = \frac{(2k-1)!!}{\sigma^{2k-1}}$$

$$E[X^{2k+1}] = 0$$

$$E[X^{2k}] = \frac{(2k-1)!!}{\sigma^{2k-1}}$$

$$E[X^{2k+1}] = 0$$

$$E[X^{2k}] = \frac{(2k-1)!!}{\sigma^{2k-1}}$$

$$E[X^{2k+1}] = 0$$

$$E[X^{2k}] = \frac{(2k-1)!!}{\sigma^{2k-1}}$$

$$E[X^{2k+1}] = 0$$

$$E[X^{2k}] = \frac{(2k-1)!!}{\sigma^{2k-1}}$$

$$E[X^{2k+1}] = 0$$

$$E[X^{2k}] = \frac{(2k-1)!!}{\sigma^{2k-1}}$$

$$E[X^{2k+1}] = 0$$

$$E[X^{2k}] = \frac{(2k-1)!!}{\sigma^{2k-1}}$$

$$E[X^{2k+1}] = 0$$

$$E[X^{2k}] = \frac{(2k-1)!!}{\sigma^{2k-1}}$$

$$E[X^{2k+1}] = 0$$

$$E[X^{2k}] = \frac{(2k-1)!!}{\sigma^{2k-1}}$$

$$E[X^{2k+1}] = 0$$

$$\begin{aligned}
 7. \quad & \begin{cases} X=a & (b>a) \\ X=-b, & EX = ap - b(1-p) \\ & = ap - b + bp = b(p-1) + ap > 0 \\ & \text{才能期望收益} \end{cases} \\
 & \text{才能期望收益}
 \end{aligned}$$

8. 随机变量X的概率函数

$$p(x) = \begin{cases} \frac{1}{3}, & -1 < x < 0 \\ \frac{1}{6}, & 0 \leq x < 4 \\ 0, & \text{其他} \end{cases}$$

$$Y = X^2 \text{ 求 } \text{Cov}(X, Y)$$

$$EXY - EX \cdot EY$$

$$EXY = EX \cdot X^2 = EX^3 = \int_{-\infty}^{+\infty} x^3 \cdot p(x) dx = \int_{-1}^0 \frac{1}{3} x^3 dx + \int_0^4 \frac{1}{6} x^3 dx$$

$$EX \cdot EY$$

$$= \int_{-\infty}^{+\infty} x p(x) dx \cdot \int_{-\infty}^{+\infty} x^2 p(x) dx$$

$$= \left(\frac{1}{3} \times \frac{1}{2} + \frac{1}{6} \times \frac{16}{3} \right) \left(\frac{1}{3} \times \frac{1}{3} + \frac{1}{6} \times \frac{64}{3} \right)$$

$$= \frac{1}{6} \times \frac{8}{3} \times \frac{1}{2} \times \left(\frac{1}{9} + \frac{16}{18} \times 64 \right) = \frac{1}{6} + \frac{16}{3} \times \frac{32}{9}$$

$$\int_{-\infty}^{+\infty} x p(x) dx = \frac{1}{3} \times \frac{1}{2} + \frac{1}{6} \times \frac{16}{3} = \frac{1}{6} + \frac{8}{9} = \frac{11}{18}$$

$$\int_{-\infty}^{+\infty} x^2 p(x) dx = \frac{1}{3} \times \frac{1}{3} + \frac{1}{6} \times \frac{64}{3} = \frac{1}{9} + \frac{32}{9} = \frac{33}{9} = \frac{11}{3}$$

$$EXY = \frac{1}{6} \times 0 - \frac{1}{6} \times \frac{1}{3} = -\frac{1}{18}$$

$$\text{Cov}(X, Y) = EXY - EX \cdot EY$$

$$= -\frac{1}{18} - \frac{11}{18} \times \frac{11}{3} = -\frac{1}{18} - \frac{121}{54} = -\frac{122}{54} = -\frac{61}{27}$$

$$= -\frac{1}{6} + \frac{4}{3} = \frac{7}{6}$$

$$9. \quad P(B) = \frac{1}{6}, \quad P(B|A) = \frac{1}{3}, \quad P(A|B) = \frac{1}{2}$$

$$X = \begin{cases} 1, & A \text{ 发生} \\ 0, & A \text{ 不发生} \end{cases} \quad Y = \begin{cases} 1, & B \text{ 发生} \\ 0, & B \text{ 不发生} \end{cases}$$

$$\text{求 } X, Y \text{ 的相关系数 } \rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{DX} \sqrt{DY}}$$

$$\text{Cov}(X, Y) = EXY - EX \cdot EY$$

$$= P(AB) - \frac{1}{4} \times \frac{1}{6} = \frac{1}{12} - \frac{1}{24} = \frac{1}{24}$$

$$= \frac{1}{24}$$

$$\sqrt{DX} = \sqrt{\frac{1}{12} - \left(\frac{1}{4}\right)^2} = \sqrt{\frac{1}{12} - \frac{1}{16}} = \sqrt{\frac{1}{48}} = \frac{1}{4\sqrt{3}}$$

$$= \frac{1}{2\sqrt{3}}$$

$$\sqrt{DY} = \sqrt{\frac{1}{6} - \left(\frac{1}{6}\right)^2} = \sqrt{\frac{1}{6} - \frac{1}{36}} = \sqrt{\frac{5}{36}} = \frac{\sqrt{5}}{6}$$

$$= \frac{\sqrt{5}}{6}$$

$$\rho_{XY} = \frac{\frac{1}{24}}{\frac{1}{4\sqrt{3}} \times \frac{\sqrt{5}}{6}} = \frac{1}{24} \times \frac{24\sqrt{15}}{\sqrt{5}} = \frac{\sqrt{15}}{5}$$

10. 设 $X_1, X_2, \dots, X_n (n \geq 2)$

$$\text{独立同分布于 } N(0, \sigma^2), \quad \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \quad Y_i = X_i - \bar{X}, \quad i=1, 2, \dots, n$$

求 Y_i 的方差 $D(Y_i), i=1, 2, \dots, n$

$$D(Y_i) = EY_i^2 - (EY_i)^2$$

$$= EY_i^2 - 0 = EY_i^2 = E(X_i - \bar{X})^2 = E(X_i^2 - 2X_i\bar{X} + \bar{X}^2)$$

$$= EX_i^2 + E\bar{X}^2 - 2EX_i\bar{X}$$

$$EX_i^2 = \sigma^2, \quad E\bar{X}^2 = \frac{\sigma^2}{n}$$

$$EX_i\bar{X} = \frac{1}{n} EX_i \sum_{j=1}^n X_j = \frac{1}{n} EX_i^2 = \frac{\sigma^2}{n}$$

$$= \sigma^2 + \frac{\sigma^2}{n} - 2 \times \frac{\sigma^2}{n} = \sigma^2 - \frac{\sigma^2}{n} = \frac{(n-1)\sigma^2}{n}$$

$$= \frac{(n-1)\sigma^2}{n}$$

$$D(Y_i) = \frac{(n-1)\sigma^2}{n}$$

$$= \frac{(n-1)\sigma^2}{n}$$

$$= \frac{(n-1)\sigma^2}{n}$$

$$EX_i\bar{X} = \frac{1}{n} EX_i \sum_{j=1}^n X_j = \frac{1}{n} EX_i^2 = \frac{\sigma^2}{n}$$

$$EX_i^2 = \sigma^2, \quad E\bar{X}^2 = \frac{\sigma^2}{n}$$

$$EX_i\bar{X} = \frac{1}{n} EX_i \sum_{j=1}^n X_j = \frac{1}{n} EX_i^2 = \frac{\sigma^2}{n}$$

$$= \sigma^2 + \frac{\sigma^2}{n} - 2 \times \frac{\sigma^2}{n} = \sigma^2 - \frac{\sigma^2}{n} = \frac{(n-1)\sigma^2}{n}$$

$$= \frac{(n-1)\sigma^2}{n}$$

$$= \frac{(n-1)\sigma^2}{n}$$

$$= \frac{(n-1)\sigma^2}{n}$$

$$= \frac{(n-1)\sigma^2}{n}$$

求 Y_i, Y_n 的相关系数 $DY_i = \frac{n-1}{n} \sigma^2$

$$\rho(Y_i, Y_n) = \frac{EY_i \cdot Y_n - EY_i \cdot EY_n}{\sqrt{DY_i} \times \sqrt{DY_n}}$$

$$EY_i \cdot Y_n = E[(X_i - \bar{X})(X_n - \bar{X})]$$

$$= EX_i \cdot X_n + E\bar{X}^2 - E(X_i + X_n)\bar{X}$$

$$= EX_i \cdot EX_n + E\bar{X}^2 - E\frac{1}{n}(X_i^2 + X_n^2)$$

$$= E\bar{X}^2 - \frac{1}{n}(EX_i^2 + EX_n^2)$$

$$= \frac{\sigma^2}{n} - \frac{2}{n}\sigma^2 = -\frac{1}{n}\sigma^2$$

$$EY_i \cdot EY_n = 0 \cdot 0 = 0$$

$$\rho(Y_i, Y_n) = \frac{-\frac{1}{n}\sigma^2}{\frac{n-1}{n}\sigma^2} = -\frac{1}{n-1}$$